

Assignment

1)

<u>Load (N)</u>	<u>Extension (mm)</u>
5	1.2
10	2.5
15	3.7
20	4.9
25	6.1

here, $x \rightarrow \text{load (N)}$
 $y \rightarrow \text{extension (mm)}$

<u>x</u>	<u>y</u>	<u>xy</u>	<u>x²</u>
5	1.2	6	25
10	2.5	25	100
15	3.7	55.5	225
20	4.9	98	400
25	6.1	152.5	625
<u>$\Sigma x = 75$</u>	<u>$\Sigma y = 18.4$</u>	<u>$\Sigma xy = 337$</u>	<u>$\Sigma x^2 = 1375$</u>

$$\text{Slope (m)} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$M = \frac{5 \times 337 - 75 \times 18.4}{5 \times 1875 - (75)^2} = \frac{305}{1250} = \underline{\underline{0.244}}$$

$$\therefore y = mx + c \Rightarrow y = \underline{\underline{0.244x + c}}$$

$$\text{Now, } c = \frac{\sum y - m \sum x}{n} = \frac{18.4 - 0.244 \times 75}{5} = \frac{0.1}{5} = \underline{\underline{0.02}}$$

$$\therefore y = \underline{\underline{0.244x + 0.02}} \Rightarrow \text{ML Model.}$$

$\therefore$ The extension of the load 55 N will be:

$$y = 0.244(55) + 0.02$$

$$y = 13.42 + 0.02 \Rightarrow y = \underline{\underline{13.44 \text{ mm}}}$$

$\therefore$ The extension will be 13.44 mm